

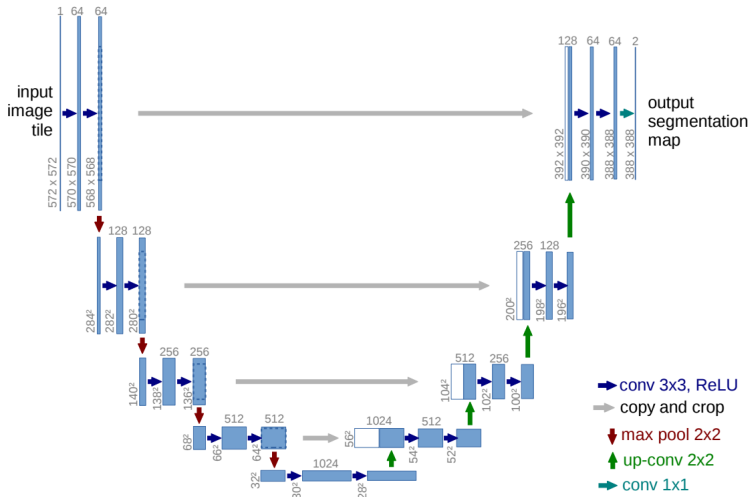
Deep Learning

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UNet



1 Kullback-Leibler Divergence

2 Autoencoders

KL Divergence

The KL divergence (also called relative entropy) is a measure of how one probability distribution is different from a second, reference probability distribution:

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but there is no symmetry, i.e. $KL(P||Q) \neq KL(Q||P)$.

Jensen-Shannon Divergence

JS Divergence is the following

$$JS(P||Q) = \frac{1}{2}KL(P||M) + \frac{1}{2}KL(M||Q),$$

where $M = \frac{P + Q}{2}$.

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Recall that probability density function (if it exists) of multivariate normal distribution with mean μ and with (non-singular, symmetric, positive definite) covariance matrix Σ is the following function:

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Suppose that we have two multivariate normal distributions:

$\mathcal{N}_1(\mu_1, \Sigma_1), \mathcal{N}_2(\mu_2, \Sigma_2)$. Then

$$KL(\mathcal{N}_1, \mathcal{N}_2) = \frac{1}{2} \left(\text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) - k + \ln \frac{|\Sigma_2|}{|\Sigma_1|} \right).$$

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In one dimensional case we will have

$$KL(\mathcal{N}_1, \mathcal{N}_2) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

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Examples?

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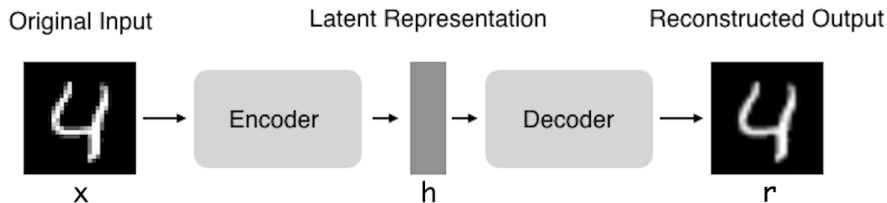
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Autoencoders



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- In a lot of different tasks.

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